

Panel MS-AR(1) with random-effects intercepts, 1970Q1 onward

Pulled real GDP per worker

1 Model

Joint panel Markov-switching AR(1) around a *linear* trend, with three regimes:

$$y_{it} = a_i + g t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}), \quad (3)$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2) \quad (\text{random effects}). \quad (4)$$

The outcome is log real GDP per worker from the pulled quarterly panel (constant 2015 USD, market FX). The trend is linear in calendar time t (g per year, common across countries). In the pooled specification $a_i \equiv a$. In the random-effects specification the a_i are i.i.d. Normal draws; α and ω are estimated by maximum likelihood, integrating each country's Hamilton-filter likelihood by 11-point Gauss–Hermite quadrature. Cycles are plotted using the posterior mean of a_i . Latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1} . σ and ρ switch with the regime. The median μ is pinned at 0. $|\rho| < 0.995$. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters. π is the ergodic distribution of Π . $E[z]$ is the long-run mean of the cycle implied by μ , ρ , and Π .

2 Common intercept and linear trend

Pulled panel from 1970Q1: log real GDP per worker. 101 countries, 7354 observations, 1970-Q1–2026-Q2. Fitted countries: 92. Observations: 6787. Log-likelihood: 15566.07. Ergodic mean of the cycle $E[z] = -0.1219$ (0.0650). Common intercept $a = 10.9411$ (0.0836). Common slope $g = 0.0021$ (0.0009).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-1.5918 (0.1302)	0.0000 —	0.3677 (0.0965)
σ	0.0809 (0.0023)	0.0249 (0.0005)	0.0108 (0.0002)
ρ	0.9950 (0.0000)	0.9950 (0.0000)	0.9950 (0.0000)
$\Pi_0.$	0.8849 (0.0153)	0.0923 (0.0142)	0.0228 (0.0063)
$\Pi_1.$	0.0201 (0.0042)	0.9603 (0.0047)	0.0196 (0.0037)
$\Pi_2.$	0.0270 (0.0040)	0.0038 (0.0034)	0.9692 (0.0037)
π	0.1690 (0.0180)	0.4305 (0.0321)	0.4004 (0.0337)

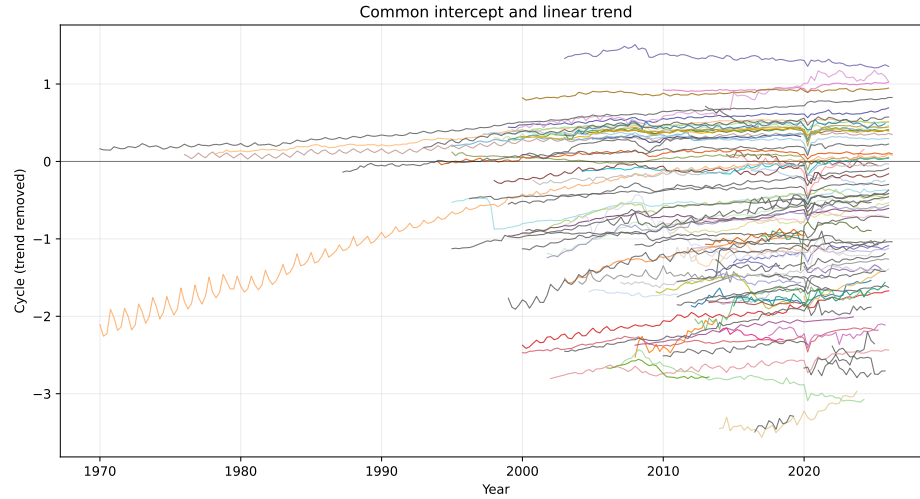


Figure 1: Country cycles after removing the linear trend.

3 Random-effects intercepts, common linear trend

Pulled panel from 1970Q1: log real GDP per worker. 101 countries, 7354 observations, 1970-Q1–2026-Q2. Fitted countries: 92. Observations: 6787. Log-likelihood: 15796.09. Ergodic mean of the cycle $E[z] = 0.0875$. Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = 9.8328$ (0.0725), $\omega = 0.7115$ (0.0206). Posterior-mean a_i : mean 9.848, min 8.040, max 11.802. Common slope $g = 0.0086$ (0.0008).

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.9859 (0.2396)	0.0000 —	0.5375 (0.0700)
σ	0.0877 (0.0029)	0.0254 (0.0005)	0.0108 (0.0002)
ρ	0.9946	0.9950 (0.0002)	0.9950
Π_0	0.8772 (0.0168)	0.0969 (0.0157)	0.0259 (0.0075)
Π_1	0.0158 (0.0033)	0.9693 (0.0041)	0.0150 (0.0030)
Π_2	0.0186 (0.0031)	0.0068 (0.0028)	0.9746 (0.0032)
π	0.1220 (0.0155)	0.4742 (0.0337)	0.4038 (0.0344)

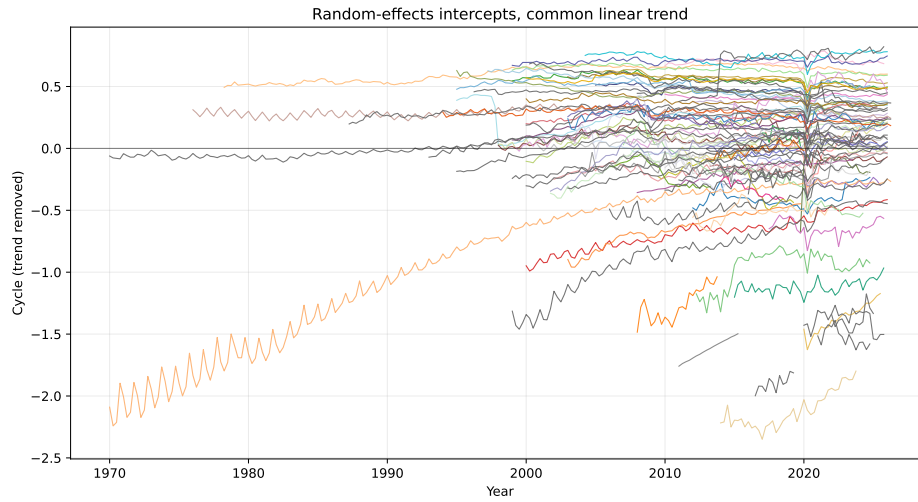


Figure 2: Country cycles after removing the linear trend.